

Danilack 2015 Study Guide Transcript

Companion reading handout for simultaneous listening.

This is the opening segment of the study guide for the paper by Danilack and colleagues, titled, The effect of labour induction on the risk of caesarean delivery: using propensity scores to control confounding by indication.

This first part is a guided oral reading of the paper itself. The goal is not to rush through the article. The goal is to hear the paper as if it were being taught by a careful clinical epidemiologist: preserving the authors' argument, but translating the dense written structure into speech that can be followed without seeing the page.

Before we begin, hold the central question in mind.

Does induction of labour increase the risk of caesarean delivery?

That sounds simple, but it is not simple. The problem is that induction is not assigned randomly in ordinary clinical practice. Patients are induced for reasons. Some of those reasons, such as pre-eclampsia, fetal growth restriction, oligohydramnios, diabetes, or post-term pregnancy, may themselves increase the chance of caesarean delivery. This is the core epidemiologic problem in the paper. It is called confounding by indication.

So as we listen, keep asking one question.

Are we seeing the effect of induction itself, or are we seeing the effect of the clinical circumstances that led to induction?

Now let us begin the oral reading.

The paper is titled, The effect of labour induction on the risk of caesarean delivery: using propensity scores to control confounding by indication.

The authors are V A Danilack, D D Dore, E W Triche, J H Muri, M G Phipps, and D A Savitz.

The paper was accepted on August thirteenth, twenty fifteen, and published online on September twenty eighth, twenty fifteen. It appeared in B J O G in twenty sixteen, volume one hundred twenty three, pages fifteen twenty one through fifteen twenty nine. The digital object identifier is ten point one one one one, slash, fourteen seventy one dash zero five two eight dot one three six eight two.

The abstract begins with the objective.

The objective was to use propensity score methods to control for confounding by indication in the association between labour induction and caesarean delivery.

Let us pause on that phrase, because it is the engine of the whole article. A propensity score is an estimated probability that a patient receives the treatment, here induction of labour, based on observed clinical and demographic features. The purpose is to make the induced and non-induced groups more comparable, at least with respect to measured variables.

The design was a cross-sectional analysis of administrative hospital discharge data, supplemented by medical record information.

The setting was fourteen United States member hospitals of the National Perinatal Information Center.

The sample was large: one hundred sixty six thousand, five hundred fifty nine singleton liveborn deliveries, from two thousand seven through two thousand twelve.

The methods were as follows. The authors used propensity scores, abbreviated P S, to balance eighty three covariates between induced and non-induced women. They compared those estimates with traditional covariate adjustment. They estimated propensity scores for labour induction versus expectant management of pregnancy, week by week, from thirty four to forty two weeks of gestation. They estimated risk ratios, abbreviated R R, for the association between labour induction and primary caesarean delivery. They did this using models with no adjustment, models with traditional adjustment for five covariates, models using matched propensity scores, and models using adjustment for the continuous propensity score.

The main outcome measure was caesarean delivery in the current or subsequent week of gestation.

The results are important. In crude models, labour induction increased the risk of caesarean delivery in all weeks, with risk ratios from one point zero six to one point five two, except at thirty nine weeks, where the risk ratio was zero point eight nine. But after matching on propensity score, the analysis showed a significantly decreased risk of caesarean delivery with labour induction during weeks thirty five through thirty nine, with risk ratios from zero point seven seven to zero point nine two. At forty weeks, the risk ratio was one point two two. At forty one weeks, it was one point three nine. Traditional covariate adjustment and propensity score adjustment gave estimates between the crude estimates and the propensity-score-matched estimates.

The conclusion of the abstract is this. There is evidence of considerable confounding by indication in the association of labour induction and caesarean delivery, particularly for preterm deliveries. Using propensity score methods, the authors found a reduced risk of caesarean delivery with labour induction before forty weeks of gestation, and an elevated risk from forty through forty two weeks.

The keywords are caesarean section, confounding variables, induced labour, obstetric delivery, pregnancy, and propensity score.

The tweetable abstract says: with confounding adjustment, labour induction does not increase the risk of caesarean at thirty four to thirty nine weeks of gestation.

Now we move to the introduction.

Labour induction is an obstetric intervention considered when there is a probable health benefit to initiating delivery rather than continuing expectant management. Examples include pre-eclampsia, post-term pregnancy, oligohydramnios, and intrauterine growth restriction.

Although both induced labour and spontaneous labour can end in caesarean delivery, there has been concern that induction may increase the likelihood of caesarean delivery. Judicious use of caesarean delivery is warranted because caesarean is major surgery. It carries increased risks for the mother in the current pregnancy, in later pregnancies, and for the newborn.

The authors then describe why the question has been difficult to answer. There has been much research on the association between labour induction and caesarean delivery, but different study designs and conflicting results have made consensus difficult.

The main reason is confounding by indication.

Confounding by indication occurs in observational studies of medical interventions when a characteristic that influences receipt of the intervention also increases the likelihood of the outcome. In this setting, women whose labour is induced may require emergency caesarean delivery because of the medical reason that led to early delivery, not because of the induction itself.

This is clinically intuitive. A woman induced for severe pre-eclampsia, fetal growth restriction, or oligohydramnios is not interchangeable with a woman who arrives in spontaneous labour without complications. If we simply compare induced labour with spontaneous labour, we may unfairly blame the induction for the risk carried by the underlying condition.

The authors then introduce propensity scores.

Propensity scores offer a more complex approach to isolating the influence of labour induction on caesarean delivery, compared with traditional analytical methods. In this study, propensity scores reflect an individual's predicted probability of being induced, versus not being induced, in a particular week, given the covariates used in the model.

Propensity scores combine many covariates into a single measure. This can allow control of more covariates than would be feasible with traditional outcome regression. When the model is specified well, exposed and unexposed subjects with similar propensity scores should have similar distributions of baseline covariates. Consequently, the effect estimates should be less confounded by baseline characteristics included in the propensity score.

The authors are careful. They do not claim that propensity scores completely resolve confounding by indication. They state that propensity scores are a statistically efficient method to address this important clinical concern, and an effective alternative to traditional observational approaches.

The introduction ends with the objective. The objective was to limit the impact of confounding by indication in the association between labour induction and caesarean delivery, using two propensity score methods: matching and adjustment.

Now we move to the methods.

The study included deliveries between January first, two thousand seven, and December thirty first, two thousand twelve, at fourteen member hospitals of the National Perinatal Information Center, also known as N P I C, slash, Quality Analytic Services.

The National Perinatal Information Center collects and analyzes administrative discharge data, supplemented by information abstracted from medical records, from member hospitals throughout the United States.

The authors included linked mother and newborn records from deliveries occurring at thirty four through forty two completed weeks of gestation.

They clarify an important dating point. Whenever the article refers to a gestational age, it means completed weeks of gestation. So thirty seven weeks means thirty seven weeks and zero days through thirty seven weeks and six days.

The authors restricted the sample to records with valid parity data. Parity was dichotomized as primipara, meaning no previous births, or multipara, meaning at least one previous birth.

They excluded stillbirths. They excluded women with prior caesarean delivery. They excluded records with maternal age outside the range of eight to sixty four years. They also excluded records with International Classification of Diseases, Ninth Revision codes, or I C D nine codes, for conditions commonly accepted as contraindications to labour induction or vaginal delivery.

The paper's flowchart begins with one million, two hundred fourteen thousand, seven hundred eighty three maternal-newborn linkages. From there, eight hundred fifteen thousand, fifty three had valid gestational age from thirty four to forty two weeks. Then two hundred twenty six thousand, five hundred ninety six had valid parity. After exclusions for maternal age outside the defined range, multifetal gestation, stillbirth, placenta previa, cephalopelvic disproportion, non-vertex presentation, prior caesarean delivery, and uterine abnormality, the final sample was one hundred sixty six thousand, five hundred fifty nine.

Labour induction was identified by I C D nine procedure codes seventy three point one, seventy three point zero one, and seventy three point four. Caesarean delivery was identified by I C D nine procedure codes seventy four point zero, seventy four

point one, seventy four point two, seventy four point four, and seventy four point nine nine.

The authors also needed to classify caesarean deliveries without labour induction as either scheduled before labour, meaning pre-labour, or occurring during labour. They adapted an algorithm based on I C D nine diagnosis codes suggestive of active labour.

Now we come to the propensity score models.

The propensity score is the probability, from zero to one, of labour induction relative to expectant management, given the covariates and membership in the study cohort.

The authors emphasize two advantages of propensity score matching for this analysis.

First, propensity scores allow for causal interpretations because matched pairs conceptually represent the same person under different treatment scenarios. That is, two patients with very similar likelihoods of being induced are compared, one induced and one not induced.

Second, propensity scores allow control of a large set of covariates, overcoming a limitation of traditional regression models. By incorporating many variables into the score, the authors hoped to account for obvious confounders, such as hypertensive and diabetic disorders, and also more subtle influences that may correlate with unmeasured clinical reasons for induction.

They used a non-parsimonious approach. In plain language, they did not try to make the model small and elegant. They tried to make it broad, because the purpose was confounding control.

They used four sources of variables.

The first source was demographic variables: maternal age, race, parity, insurance, hospital, and year of discharge.

The second source was the one hundred most common I C D nine diagnosis codes, up to the fifth digit, among deliveries from thirty four through forty two weeks. They excluded codes for conditions arising after delivery, and conditions that might first be diagnosed after the onset of labour.

The third source was conditions hypothesized as possible confounders. These included intrauterine growth restriction, placenta abruption, pre-gestational hypertension, gestational hypertension, pre-eclampsia superimposed on pre-existing hypertension, mild pre-eclampsia, severe pre-eclampsia, eclampsia, pre-gestational diabetes types one and two separately, and gestational diabetes.

The fourth source was conditions on the Joint Commission list of conditions that may justify elective delivery before thirty nine weeks, if those conditions were not already included.

Next, the authors describe the week-by-week design.

They conducted analyses of successive assignments of labour induction versus expectant management. For each week from thirty four to forty two weeks, the induction group included women induced in that week. The expectant management group included women with spontaneous labour in that same week, excluding pre-labour caesarean deliveries in that week, and also included all deliveries in subsequent weeks, regardless of later method of delivery or onset of labour.

This is a crucial design choice. The comparator is not simply spontaneous labour. The comparator is expectant management. In clinical decision-making, the real alternative to induction this week is not spontaneous labour this week. The real alternative is waiting, with whatever happens afterward.

For each week, the authors calculated propensity scores for labour induction using logistic regression with demographic, comorbidity, and pregnancy complication variables as predictors. For all models, selected variables were removed until the models converged and did not have linear combinations or extreme standard errors.

Now we move to propensity score matching.

The authors performed one-to-one matching between labour inductions and expectantly managed pregnancies, using the propensity score for each week of gestation.

The matching algorithm first tried to match on five digits of the propensity score. For remaining unmatched cases, it tried four digits, then fewer digits, until a one-digit match. The maximum allowed difference in propensity score between matched patients was zero point one.

The authors evaluated the matched data sets for overlap of propensity scores and used standardized differences to assess covariate balance between the induced and expectantly managed groups.

Now we move to outcome models.

In the full analytic sample, the authors used log-binomial regression to model primary caesarean delivery based on labour induction status. This gave crude, unadjusted risk ratios for each week.

They used log-binomial models because odds ratios from logistic regression would be expected to overestimate the risk of caesarean delivery, since caesarean delivery is not a rare outcome.

They repeated the regression models with traditional covariate adjustment for hypertensive disorders, diabetic disorders, intrauterine growth restriction, maternal age, and parity.

Among the propensity-score-matched deliveries, they used a matched-pair analysis and regressed caesarean delivery on labour induction. They did this with and without adjustment for additional characteristics that remained unbalanced after matching.

For the propensity score adjustment approach, the propensity score was included as a continuous covariate in the log-binomial regression model. Observations outside overlapping regions of propensity score were excluded. Modified Poisson regression was used when negative binomial regression models did not converge.

The statistical analysis was conducted using SAS version nine point three. PROC LOGISTIC was used to calculate propensity scores, and PROC GENMOD was used for the log-binomial and modified Poisson models. The study had institutional review board approval and exemption as described by the authors.

Now we move to the results.

The final sample included one hundred sixty six thousand, five hundred fifty nine mother-newborn dyads from fourteen hospitals.

After excluding pre-labour caesarean deliveries at week thirty four that did not contribute to the analyses, the starting analytic sample was one hundred sixty six thousand, four hundred forty one.

The percentage of eligible women induced each week ranged from zero point three percent at thirty four weeks to fifty nine percent at forty two weeks.

The overall primary caesarean delivery rate, after excluding pre-labour caesareans at thirty four weeks, was seventeen percent.

The propensity score models were large. The number of variables incorporated into the week-specific models was seventy six variables at thirty four weeks, eighty at thirty five weeks, eighty two at thirty six weeks, eighty three at thirty seven weeks, eighty at thirty eight weeks, eighty two at thirty nine weeks, eighty one at forty weeks, seventy five at forty one weeks, and fifty one at forty two weeks.

For weeks thirty four through thirty eight, more than ninety nine percent of inductions were matched to an expectantly managed pregnancy. At weeks thirty nine through forty one, the match rate was between eighty one and ninety percent. At forty two weeks, the match rate was lower, fifty eight percent.

The authors examined covariate balance across eighty three considered covariates. The number of unbalanced covariates after matching was small: five at thirty four weeks, four at thirty five weeks, five at thirty six weeks, four at thirty seven weeks, six at thirty eight weeks, four at thirty nine weeks, three at forty weeks, none at forty one weeks, and three at forty two weeks.

Now let us listen to the first major table in prose.

The first table describes propensity score matching. At thirty four weeks, the starting sample was one hundred sixty six thousand, four hundred forty one. There were four hundred forty one inductions that week. The matched data set had eight hundred seventy four women. Ninety nine point one percent of inductions were matched. The matched propensity score risk ratio for caesarean delivery was zero point seven nine,

with a confidence interval from zero point six three to one point zero zero. After adjustment for remaining unbalanced covariates, the risk ratio was zero point seven eight, with a confidence interval from zero point six two to zero point nine nine six.

At thirty five weeks, there were six hundred twenty five inductions. Ninety nine point four percent were matched. The matched risk ratio was zero point seven seven, with a confidence interval from zero point six five to zero point nine one. After additional adjustment, it was zero point seven four, with a confidence interval from zero point six two to zero point eight seven.

At thirty six weeks, there were one thousand, three hundred fifty one inductions. Ninety nine point eight percent were matched. The matched risk ratio was zero point seven eight, with a confidence interval from zero point six eight to zero point eight nine. After additional adjustment, it was zero point seven seven, with a confidence interval from zero point six eight to zero point eight eight.

At thirty seven weeks, there were three thousand, eight hundred eighty one inductions. Ninety nine point eight percent were matched. The matched risk ratio was zero point seven nine, with a confidence interval from zero point seven three to zero point eight six. After adjustment, it remained zero point seven nine.

At thirty eight weeks, there were seven thousand, eighty three inductions. Ninety nine point seven percent were matched. The matched risk ratio was zero point eight three, with a confidence interval from zero point seven seven to zero point eight nine. After adjustment, it was zero point eight one, with a confidence interval from zero point seven five to zero point eight six.

At thirty nine weeks, the sample becomes clinically very large. There were nineteen thousand, six hundred fifty five inductions. Eighty two point four percent were matched. The matched risk ratio was zero point nine two, with a confidence interval from zero point eight seven to zero point nine seven. After adjustment, it was zero point nine zero, with a confidence interval from zero point eight six to zero point nine five.

At forty weeks, there were twelve thousand, one hundred fifteen inductions. Eighty nine point nine percent were matched. The matched risk ratio was one point two two, with a confidence interval from one point one five to one point two nine. After adjustment, it was one point two three.

At forty one weeks, there were five thousand, eight hundred nine inductions. Eighty one point one percent were matched. The matched risk ratio was one point three nine, with a confidence interval from one point two eight to one point five one. Adjustment did not change it.

At forty two weeks, there were three hundred sixty seven inductions. Only fifty seven point eight percent were matched. The matched risk ratio was one point zero five, with a confidence interval from zero point seven eight to one point four three. After adjustment, it was still about one point zero five.

Here is the table's message in words. In the propensity-score-matched analysis, induction from thirty four through thirty nine weeks was not associated with a higher caesarean risk. It was associated with a lower caesarean risk. At forty and forty one weeks, induction was associated with a higher caesarean risk. At forty two weeks, the estimate was not clearly elevated, but the sample and match rate were limited.

Now let us hear the second table.

The second table compares crude risk ratios, traditionally adjusted risk ratios, and propensity-score-matched risk ratios.

At thirty four weeks, the crude risk ratio was one point two eight. Traditional adjustment brought it down to zero point eight two. Propensity score matching brought it to zero point seven nine.

At thirty five weeks, the crude risk ratio was one point four zero. Traditional adjustment brought it down to zero point eight nine. Propensity score matching brought it to zero point seven seven.

At thirty six weeks, the crude risk ratio was one point two five. Traditional adjustment was zero point eight nine. Propensity score matching was zero point seven eight.

At thirty seven weeks, the crude risk ratio was one point one seven. Traditional adjustment was zero point eight nine. Propensity score matching was zero point seven nine.

At thirty eight weeks, the crude risk ratio was one point zero six. Traditional adjustment was zero point nine six. Propensity score matching was zero point eight three.

At thirty nine weeks, the crude risk ratio was zero point eight nine. Traditional adjustment was one point zero zero. Propensity score matching was zero point nine two.

At forty weeks, the crude risk ratio was one point three nine. Traditional adjustment was one point three seven. Propensity score matching was one point two two.

At forty one weeks, the crude risk ratio was one point five one. Traditional adjustment was one point four three. Propensity score matching was one point three nine.

At forty two weeks, the crude risk ratio was one point five two. Traditional adjustment was one point two six. Propensity score matching was one point zero five.

The trend is the teaching point. In eight of the nine weeks, the crude estimate was larger than the traditionally adjusted estimate, and the traditionally adjusted estimate was larger than the propensity-score-matched estimate. That pattern strongly suggests that confounding by indication was pushing the crude association upward.

Now let us hear the third table, which describes propensity score adjustment rather than matching.

The authors trimmed observations outside the overlapping propensity score regions. Across weeks, this removed between zero and nine percent of total observations.

At thirty four weeks, after trimming, the log-binomial model did not converge, but the modified Poisson risk ratio was zero point nine seven.

At thirty five weeks, the modified Poisson risk ratio was one point zero four.

At thirty six weeks, it was zero point nine seven.

At thirty seven weeks, the log-binomial and modified Poisson estimates were both about zero point nine one.

At thirty eight weeks, the risk ratio was about zero point eight six to zero point eight seven.

At thirty nine weeks, it was zero point nine four.

At forty weeks, it was one point two six.

At forty one weeks, it was about one point three five to one point three six.

At forty two weeks, it was about one point two zero.

The authors summarize this by saying that propensity score adjustment gave estimates closer to the null than propensity score matching for weeks thirty four through thirty nine and week forty one, but not for weeks forty or forty two.

Now we move to the discussion.

The main finding is that advanced confounding adjustment suggests considerable confounding by indication in the association between labour induction and caesarean delivery. This confounding appears especially substantial before thirty nine weeks.

The crude results suggested that induction was generally associated with increased primary caesarean risk. But after controlling for demographic factors, maternal conditions, and pregnancy complications through propensity scores, induction was associated with reduced risk at thirty four through thirty nine weeks, and elevated risk only at forty through forty two weeks.

The authors then explain the meaning of the two propensity score approaches.

In propensity score matching, the estimated association is the effect of labour induction among women who were actually induced. This is sometimes called the effect among the treated. Clinically, this matters because it asks: among women who are induced in real practice, what happens when they are compared with similar women who are managed expectantly?

By contrast, including the propensity score as a covariate in the regression model estimates a more traditional population-average association. That asks about the risk among all women empirically eligible for induction or expectant management.

The larger effect estimates with propensity score adjustment, compared with propensity score matching, suggest that women who were actually induced differed from the overall population. It also suggests that residual confounding may be more likely in one of the models.

The authors interpret their results this way. From thirty four through thirty nine weeks, the act of inducing labour does not appear to increase caesarean risk. From forty through forty two weeks, a higher risk of caesarean after induction cannot be ruled out.

For thirty four through thirty nine weeks, the findings may suggest that when there is a reason to deliver, inducing labour may be more effective for achieving vaginal delivery than waiting. Confounding by indication may be more obvious at earlier gestational ages, and the authors may have had better ability to adjust for the indications that lead to intervention before forty weeks.

For forty through forty two weeks, several interpretations are possible. Induction may truly increase caesarean risk. There may be residual confounding by indication. Or the finding may be affected by truncating the dataset at forty two weeks, because at forty two weeks the expectant management group includes only spontaneous deliveries at forty two weeks.

The authors add important clinical considerations. The longer labour curve for induced labour may be less likely to be tolerated in late-term or post-term pregnancy. Conditions such as oligohydramnios, macrosomia, and meconium-stained fluid become more common later in gestation and may increase the chance of caesarean. The authors attempted to control for oligohydramnios. Excessive fetal growth may not have been an ideal proxy for macrosomia. They did not control for meconium staining, because meconium may be identified downstream of induction.

Now we move to strengths and limitations.

The strengths include the large sample size. The large sample allowed week-specific associations and large, non-parsimonious propensity score models. The authors could include specific disorders separately. For example, hypertensive disorders were not collapsed into one broad variable; they were included as separate indicators where possible.

Now the limitations.

Unlike randomized studies, where both measured and unmeasured covariates tend to balance across treatment groups, propensity score methods in observational studies can only reduce bias from measured variables included in the propensity score.

Unmeasured variables correlated with measured variables may also become partly balanced, depending on the strength of the correlation. But unmeasured confounding remains possible.

The data came from hospital discharge records, so the exact timing of onset of conditions or complications was unknown. The authors tried to exclude variables that may be caused by induction or may occur after induction or expectant management. They assumed that included variables were baseline confounders, not time-dependent confounders affected by prior treatment.

The administrative data also lacked some important clinical detail. For example, the authors lacked information on body mass index and cervical dilatation. This matters clinically because an unfavorable cervix is a major determinant of induction course and caesarean risk.

The authors argue, however, that more precise covariate control would be expected to move the estimates further in the same direction they shifted with propensity score adjustment.

Now we move to interpretation.

The authors used propensity score methods to address confounding by indication in the relationship between labour induction and caesarean delivery. In any non-experimental study, prognostic characteristics can differ between treated and untreated subjects. This is very true with labour induction.

Previously studied confounders include maternal age, race, parity, epidural analgesia, insurance status, education, year of delivery, body mass index, hypertensive disorders, diabetic disorders, smoking, and whether there is a medical indication for delivery.

The authors argue that restriction and adjustment for selected conditions may be insufficient for complete confounding control. Propensity scores may offer a more sophisticated and effective approach.

They note that propensity score matching has been shown to produce a more balanced baseline distribution of covariates than other propensity score methods, and therefore may better approximate randomized controlled trial results.

They compare their findings with two meta-analyses of randomized trials. One meta-analysis of one hundred fifty seven randomized controlled trials reported an overall relative risk of zero point eight eight for induction versus expectant management and caesarean delivery. Another meta-analysis of thirty one randomized controlled trials reported an odds ratio of zero point eight three.

The authors' findings for preterm and term deliveries do not fully agree with these meta-analyses. However, after adjustment for confounding, their estimates moved closer to the randomized-trial estimates. They also note that similar findings have been reported by Caughey and colleagues, who found lower caesarean risk in the induction group at thirty eight and thirty nine weeks, and Rasmussen and Rasmussen, who found lower adjusted odds than crude odds at thirty nine through forty one weeks.

Now we come to the conclusion.

The authors conclude that crude associations between labour induction and caesarean delivery are confounded by factors that are common causes of both induction and caesarean delivery.

After accounting for potential confounding through propensity score matching, the act of inducing labour does not appear to cause primary caesarean delivery in women induced before forty weeks of gestation.

Pregnant women and providers concerned about caesarean risk with induction, and possibly considering pre-labour caesarean delivery as an alternative, should be informed that the risk of caesarean after induction may not be increased until forty weeks of gestation.

Nonetheless, after accounting for potential confounding at forty through forty two weeks, the risk of caesarean delivery after induction versus expectant management is only moderately elevated, approximately twenty to forty percent.

The additional information states that disclosure of interests is available online. The authors are accountable for all aspects of the research. V A Danilack contributed to conception and design, analysis, drafting, and final approval. The other authors contributed to design, interpretation, review, acquisition of data, and final approval according to their roles.

The study was approved by the Women and Infants Hospital Institutional Review Board, project thirteen dash zero zero three nine, and was determined exempt from further review by the Brown University Institutional Review Board.

There was no funding and no acknowledgements.

Supporting information included an I C D nine table for active labour, variables included in the propensity score models, balance of covariates between matched observations, and supplemental methods.

That completes the oral reading portion of the paper.

Let us close this first segment with the essential scaffold for memory.

First, the naive comparison makes induction look more dangerous than it may be, because induced patients are often sicker or more complicated to begin with.

Second, the key methodological move is to compare induction with expectant management, week by week, and to use propensity scores to balance many measured baseline covariates.

Third, after propensity score matching, induction before forty weeks does not appear to increase primary caesarean risk in this dataset. In fact, from thirty four through thirty nine weeks, the matched estimates are below one.

Fourth, after forty weeks, the signal changes. At forty and forty one weeks, induction is associated with a moderately elevated caesarean risk, but residual confounding and the structure of the late-term comparison group remain important caveats.

Now that we have read the paper, we are ready to slow down and understand its statistical heart: propensity scores.

Deep explanation of propensity scores

Let us now step away from obstetrics for a moment and talk about causal inference.

The paper by Danilack and colleagues depends on one central idea. When we compare outcomes between two treatment groups in an observational study, the two groups may not be comparable at baseline. If they are not comparable, then a simple outcome comparison can mislead us.

In a randomized controlled trial, treatment assignment is random. Randomization is powerful because it tends to distribute baseline characteristics evenly between treatment groups. Some of those characteristics are known, such as age, parity, diabetes, hypertension, or cervical dilation. Some are unknown, or not measured, or not even yet discovered. If randomization succeeds, the two groups are similar enough that a difference in outcome can be more credibly attributed to the intervention.

But in clinical practice, treatment is not random.

Patients receive treatments for reasons.

That sentence is the beginning of almost every difficult observational study.

A patient receives partial breast irradiation rather than whole breast irradiation for reasons. A patient receives induction of labour rather than expectant management for reasons. A patient receives one antibiotic, one surgical approach, or one monitoring strategy rather than another for reasons. Some of those reasons are recorded in the chart. Some are implicit in the clinician's judgment. Some are biological features that we did not measure.

If those reasons are also related to the outcome, we have confounding.

Confounding means that the measured association between treatment and outcome is distorted by another factor. The confounder is associated with the treatment choice, and it is also associated with the outcome.

In the Danilack paper, the treatment is labour induction. The outcome is primary caesarean delivery. A confounder might be severe pre-eclampsia. Severe pre-eclampsia can make induction more likely, because delivery may be indicated. It can also make caesarean delivery more likely, because the maternal and fetal condition

may deteriorate or labour may be less well tolerated. If we ignore severe pre-eclampsia, we may mistakenly attribute its caesarean risk to induction itself.

This special version of confounding is called confounding by indication.

Confounding by indication means that the very reason a patient was selected for treatment also affects the outcome. The indication for treatment becomes part of the bias.

This is why crude comparisons can be deeply unfair. Comparing induced patients with all non-induced patients is not like comparing identical patients who differ only by induction. It may be comparing higher-risk pregnancies with lower-risk pregnancies, and then blaming the higher-risk outcomes on induction.

Now let us define the propensity score.

A propensity score is the probability that a patient receives a particular treatment, given that patient's observed baseline characteristics.

Say that again in a more clinical way.

The propensity score asks: based on what we know about this patient before treatment, how likely was this patient to receive the treatment?

In Danilack, the question becomes: based on this patient's gestational age week, demographics, hospital, year, parity, and recorded maternal and pregnancy conditions, how likely was she to be induced rather than managed expectantly?

The score ranges from zero to one.

A score near zero means the patient looked very unlikely to receive the treatment. A score near one means the patient looked very likely to receive the treatment. A score near zero point five means the patient could plausibly have gone either way, based on observed characteristics.

The usual way to estimate this score is logistic regression. But notice what is being modeled. We are not yet modeling the outcome. We are not asking: who had a caesarean? We are asking: who got induced?

That is the first conceptual turn.

Traditional regression usually starts with the outcome. It asks whether induction predicts caesarean delivery after adjusting for confounders.

Propensity score analysis starts with treatment assignment. It asks whether the observed baseline characteristics predict induction. Once we know each patient's predicted probability of induction, we can use that score to create a fairer comparison.

This is a very elegant idea.

Many baseline variables are compressed into one balancing score. Instead of trying to adjust for dozens of covariates directly in an outcome model, the analyst first summarizes the treatment-selection process. Patients with similar propensity scores should have similar distributions of measured baseline covariates, even if one received treatment and the other did not.

This does not mean that two patients with the same propensity score are identical.

It means that, in aggregate, treated and untreated patients with similar scores should be balanced with respect to the measured variables used to create the score.

Let us make this concrete.

Imagine two pregnant patients at thirty nine weeks.

Patient A is induced. Patient B is not induced that week and is managed expectantly. If patient A has a propensity score of zero point eight three zero, and patient B has a propensity score of zero point eight three five, then both patients looked highly likely to be induced based on measured baseline characteristics. One actually was induced. The other was not. That pair is more informative than comparing patient A with someone whose propensity score was zero point zero five, meaning someone who looked nothing like the kind of patient who would be induced.

The point is not that the score has clinical magic. The point is that it helps compare like with like.

Now let us ask how the propensity score model is built.

The analyst chooses baseline variables that may influence treatment choice, outcome risk, or both. In an ideal analysis, these variables are chosen using clinical knowledge and causal reasoning, not just by asking which variables are statistically significant.

This matters.

A variable can be clinically important even if it is not statistically significant in the treatment-assignment model. A variable can also be correlated with other predictors and still be useful. The goal is not to build the prettiest prediction model. The goal is to create balance between treatment groups.

This is why the phrase garbage in, garbage out is so important. If the propensity score model omits crucial confounders, the final analysis may still be biased. If the model includes variables measured after treatment, it may adjust for consequences of the treatment and create new problems. If the model is poorly specified, the score may give a false sense of rigor.

In Danilack, the authors used a deliberately broad, non-parsimonious model. They included demographics, hospital, year, parity, common diagnosis codes, major obstetric conditions, and Joint Commission conditions that may justify delivery before thirty nine weeks. This is exactly the logic of propensity scores: model the reasons induction happens, as completely as the data allow.

Now, once every patient has a propensity score, what do we do with it?

There are four common approaches.

The first is matching.

The second is stratification.

The third is inverse probability of treatment weighting.

The fourth is covariate adjustment.

Let us take them one at a time, slowly.

Propensity score matching is the easiest to understand and often the most intuitive.

In matching, each treated patient is paired with one or more untreated patients who have similar propensity scores. If the treated patient had a high probability of treatment, the matched untreated patient should also have had a high probability of treatment. If the treated patient had an intermediate probability, the match should also be intermediate.

The most common version is one-to-one matching. One treated patient is matched to one control patient. But many-to-one matching is also possible, where one treated patient is matched with more than one control patient.

The usual matching method is nearest neighbor matching. This means the algorithm looks for the patient in the other group whose propensity score is closest, within a prespecified allowable distance.

That allowable distance is important. If the permitted difference is too wide, the matches may not really be comparable. If it is too narrow, many patients may remain unmatched.

Matching may be done without replacement or with replacement. Without replacement means that once a control patient is used as a match, that control patient cannot be used again. With replacement means a strong control match can be used in more than one matched pair.

After matching, the analyst checks whether the baseline characteristics are balanced between groups. This is analogous to examining Table 1 in a randomized trial.

But the key diagnostic is not the p value.

The key diagnostic is balance.

One common measure is the standardized difference. A standardized difference less than zero point one is often used as a rough sign that the mean or proportion of a baseline characteristic is similar between groups. Graphical checks may also be used, such as boxplots or quantile-quantile plots, because balance is not only about averages. The whole distribution can matter.

Matching has a major strength. It creates a study population in which the treated and untreated groups can look much more comparable.

But it also has a major cost. Unmatched patients are excluded.

This can reduce statistical power. It can also change the estimand, meaning the exact causal question being answered. Matching often estimates the treatment effect among patients who were treated and could be matched to similar untreated patients. It may not estimate the effect in every patient in the original population.

This distinction is crucial in Danilack.

When Danilack used propensity score matching, the result was most naturally interpreted as the effect of induction among women who were actually induced and had comparable expectantly managed counterparts. It is an estimate among the treated, not necessarily a universal estimate for all pregnant patients.

Now let us move to stratification.

Stratification is like a coarser version of matching.

Instead of pairing individual patients, the analyst divides the whole cohort into strata based on propensity score. A common choice is five groups, or quintiles. One group contains patients with the lowest propensity scores. Another contains the next range. This continues up to the group with the highest propensity scores.

Within each stratum, patients have propensity scores in a similar range. Therefore, in principle, the treated and untreated patients within that stratum should have a more similar distribution of measured baseline characteristics than the original untreated and treated groups overall.

The analyst then compares outcomes within each stratum. After that, the stratum-specific treatment effects are pooled into an overall estimate, often as a weighted average.

The advantage is that stratification can include all patients. It is also fairly easy to explain.

The limitation is that the groups are not as granular as matched pairs. If confounding is strong, five strata may not be enough to balance patients well. Also, if there are few outcome events, splitting the data into several strata can make estimates unstable.

Now let us move to inverse probability of treatment weighting.

This is more abstract, but it is powerful.

In inverse probability weighting, each patient receives a statistical weight. The weight is based on how likely the patient was to receive the treatment they actually received.

If a treated patient had a high probability of being treated, that patient receives a smaller weight. If a treated patient had a low probability of being treated, that patient

receives a larger weight.

For treated patients, the basic weight is the reciprocal of the propensity score. For control patients, the basic weight is the reciprocal of one minus the propensity score.

Let us translate that.

If a patient received induction and her propensity score was zero point eight, then she was fairly likely to be induced. Her weight would be one divided by zero point eight, or one point two five.

If another patient received induction but her propensity score was zero point two, she received a treatment that was unlikely based on her measured characteristics. Her weight would be one divided by zero point two, or five. She contributes more heavily to the weighted analysis because she represents a rarer type of treatment assignment.

The goal is to create a synthetic population in which treatment assignment is independent of measured baseline confounders. In that synthetic population, the distribution of measured covariates should be balanced between treatment groups.

The advantage is that all patients can be included.

The limitation is that extreme weights can make the analysis unstable. Extreme weights happen when patients received a treatment that their baseline characteristics made very unlikely. If a propensity score is very close to zero or very close to one, the reciprocal can become very large. A few patients can then dominate the analysis.

This is why overlap matters.

If the treated and untreated groups have very different propensity score distributions, the analysis may be trying to compare patients who have no real counterpart in the other group.

Think of a radiation oncology example. If every patient receiving partial breast irradiation is older than seventy, and every patient receiving whole breast irradiation is younger than seventy, then age cannot be balanced well. The data contain almost no real-world overlap. A statistical method cannot create clinical comparability out of nothing.

The same applies to obstetrics. If all induced patients at a certain gestational age have severe medical complications, and all expectantly managed patients are healthy, the propensity score ranges may not overlap enough to support a credible comparison.

Now let us move to covariate adjustment using the propensity score.

This method puts the propensity score into a conventional outcome regression model as a predictor, along with the treatment indicator.

In Danilack terms, the model asks whether induction predicts caesarean delivery while adjusting for the propensity score. The propensity score carries information about many baseline variables in one number.

This approach is straightforward and keeps all patients, or at least all patients within the region of overlap if trimming is used.

But it has limitations.

If the propensity score alone does not create enough balance, the analyst may also adjust for additional baseline variables in the outcome model. Some people call this a doubly robust style of analysis when both the propensity model and the outcome model are used to protect against misspecification.

But this reintroduces the burden of multivariable outcome modeling. If there are many predictors and few outcome events, the model can overfit.

Overfitting means the model is trying to learn too much from too little outcome information. The estimates can become unstable, inaccurate, or fail to converge. This is one reason propensity score methods are attractive in settings with many confounders but relatively few outcome events.

Now pause and summarize the four methods.

Matching pairs treated and untreated patients with similar scores. It is intuitive and rigorous, but discards unmatched patients.

Stratification groups patients by score ranges. It keeps more patients, but is coarser.

Inverse probability weighting creates a synthetic balanced population using weights. It keeps patients, but can be unstable when weights are extreme.

Covariate adjustment puts the propensity score into the outcome model. It is simple, but may not fully balance covariates and may drift back into the problems of ordinary multivariable regression.

Now let us ask why propensity scores can be better than simply adding covariates to a regression model.

Traditional multivariable regression asks the outcome model to do everything. It must estimate the treatment effect and adjust for many covariates at the same time.

This can work well when the outcome is common, the sample is adequate, and the model is not overloaded.

But when there are many confounders and few outcome events, the model can become fragile. This is especially relevant in studies where outcomes are rare, such as breast local recurrence in ductal carcinoma in situ, or in subgroup analyses where event counts shrink.

Propensity scores separate the design problem from the outcome problem.

First, we design a more comparable observational study by balancing baseline variables. Then we analyze the outcome in that more comparable sample or weighted population.

This is why many statisticians describe propensity score analysis as a design tool, not merely an analysis trick.

It asks the investigator to confront comparability before looking at the outcome.

Now let us connect this back to Danilack.

Danilack and colleagues did not simply ask: among women who were induced, how many had caesareans, compared with women who were not induced?

They asked a better clinical question.

At each week from thirty four through forty two weeks, if a woman was induced, what would be the comparable alternative? The alternative is expectant management, meaning not inducing this week and allowing the pregnancy to continue with whatever occurs later.

Then they estimated the probability of being induced in that week based on many measured baseline features.

Then they used that probability to compare induced women with expectantly managed women who had similar measured likelihoods of induction.

That is why the crude and matched results diverged.

The crude analysis said induction often looked associated with higher caesarean risk. But once women were compared with more similar women, the apparent risk decreased. Before forty weeks, the matched estimates actually moved below one.

That movement is not just a statistical detail. It is the paper's main lesson.

The apparent harm of induction may have been partly, or largely, the risk profile of the patients selected for induction.

Now let us discuss the merits of propensity scores in a balanced way.

The first merit is improved internal validity in observational studies. Propensity scores help reduce bias from measured confounders when randomization is not feasible.

The second merit is transparency. After matching or weighting, the analyst can show whether baseline characteristics are balanced. This makes the comparison more inspectable.

The third merit is efficiency when there are many confounders. Instead of placing a long list of covariates into an outcome model, the analyst can summarize treatment-selection probability in one score.

The fourth merit is clinical realism. Propensity score methods can be used in retrospective data where randomized trials are impractical, too expensive,

underpowered, unethical, or unlikely to be performed.

Now let us discuss limitations, because this is where scientific rigor matters.

The first limitation is that propensity scores adjust only for measured and included variables.

This cannot be overstated.

A propensity score cannot balance what it cannot see.

If body mass index is missing, the propensity score cannot directly balance body mass index. If cervical dilation is not recorded, the propensity score cannot directly balance cervical dilation. If a clinician's judgment about fetal reserve is not captured in a code, the propensity score cannot directly balance that judgment.

It may indirectly balance some unmeasured variables if they are correlated with measured variables. But that is not guaranteed.

This is why propensity scores do not replace randomization.

Randomization can balance known and unknown confounders in expectation.

Propensity scores can only address observed confounders, and only if the model is appropriate.

The second limitation is overlap.

There must be patients in both treatment groups with similar propensity scores. If the distributions do not overlap, the comparison is not supported by the data.

This is sometimes called the positivity assumption. For a given set of baseline characteristics, there must be a positive probability of receiving either treatment. If one type of patient always gets treatment and another type never gets treatment, the data cannot tell us what would have happened under the alternative.

In clinical language, you need patients who could plausibly have been treated either way.

The third limitation is model specification.

The propensity score model must include the right baseline variables and capture the relevant structure of treatment assignment. Poor variable choice, missing data, incorrect functional forms, or inclusion of post-treatment variables can all compromise the analysis.

The fourth limitation is that each propensity score method has its own tradeoffs.

Matching can discard patients. Stratification can be too coarse. Weighting can be unstable with extreme weights. Covariate adjustment can fail to achieve balance and can reintroduce outcome-model overfitting.

The fifth limitation is interpretability.

Different methods may estimate different causal quantities. Matching may estimate the effect among treated patients who were matchable. Weighting may estimate an average treatment effect in a synthetic population. Covariate adjustment may estimate a population-average association under model assumptions. These are related, but not identical.

Now let us make the idea memorable.

A propensity score is not the risk of the outcome.

It is the probability of treatment.

In Danilack, it is not the probability of caesarean. It is the probability of induction.

That distinction is essential.

The score is built before outcome comparison. It is a way of asking: who looked like they were going to be induced?

Once we know that, we can compare induced and non-induced patients who had similar chances of being induced.

Now imagine the clinical decision in front of you.

A patient asks, will induction increase my chance of caesarean?

If you answer from a crude observational comparison, you may say yes, because induced patients have more caesareans.

But that answer may be contaminated by confounding by indication.

A more careful answer asks: compared with patients like you, at this gestational age, with similar measured risk factors and similar reasons one might choose induction, what happened when labour was induced rather than managed expectantly?

That is the kind of question propensity scores are designed to approach.

They do not make the observational study randomized.

They make the observational comparison more disciplined.

Now let us close this section with a rigorous summary.

Propensity scores were introduced by Rosenbaum and Rubin as a way to reduce bias in observational studies of treatment effects.

The score is the estimated probability of receiving treatment, conditional on observed baseline characteristics.

The analyst can use the score through matching, stratification, inverse probability weighting, or covariate adjustment.

The goal is to balance measured baseline covariates between treatment groups, so that the comparison resembles, in limited ways, the baseline comparability of a

randomized trial.

The method is strongest when the dataset is large, baseline variables are well measured, the propensity model is clinically and statistically well specified, and there is substantial overlap in propensity scores between treatment groups.

The method is weakest when key confounders are unmeasured, data are missing, treatment groups do not overlap, or the method is treated as a statistical cure-all.

And the most important sentence is this.

Propensity scores can reduce confounding from what we measured, but they cannot rescue us from what we failed to measure.

That is the scientific humility we must carry back into the Danilack paper.

The study's conclusion is not that propensity scores prove induction is harmless. The conclusion is more precise. In this large observational dataset, after balancing many measured factors associated with induction, the association between induction and primary caesarean delivery shifted substantially. Before forty weeks, induction no longer appeared to increase caesarean risk. At forty and forty one weeks, a moderate elevation remained.

That is a much better, more careful answer than the crude comparison.

Propensity scores, the mathematical layer

Now we are ready for a more mathematical explanation.

This section is challenging in audio, because causal inference is often taught with notation. But we do not need to hear symbols in order to understand the mathematics. We need to hear the structure.

So I will translate the notation into spoken language.

The first idea is this.

A propensity score is a dimension-reduction tool for causal inference.

That sounds abstract. Let us unpack it.

Each patient has many baseline characteristics. In Danilack, these include gestational age week, parity, maternal age, hospital, year, insurance, hypertensive disorders, diabetic disorders, growth restriction, abruption, and many diagnosis-code variables. Mathematically, we can imagine all of those variables as a long list attached to each patient.

That long list is high-dimensional.

If there are eighty baseline variables, then every patient is described by an eighty-dimensional profile. To compare induced and non-induced patients exactly on all

eighty variables would be nearly impossible. There may be no two patients who match exactly. This is the curse of dimensionality.

The propensity score compresses that long baseline profile into one number: the probability that the patient receives treatment.

That is the dimension reduction.

Many covariates go in. One probability comes out.

But this is not arbitrary compression. It is compression with a causal purpose. The score is designed to preserve the information needed to balance measured treatment-selection factors between groups.

Now let us build the formal causal framework in words.

The framework is called the Neyman-Rubin causal model, also known as the potential outcomes framework.

Imagine a study with many subjects. For each subject, we define treatment assignment. Treatment assignment has two possibilities. One means the subject receives treatment. Zero means the subject receives control.

In Danilack, treatment equals induction in a given gestational week. Control equals expectant management in that week.

Each subject also has baseline covariates. These are the measured features that exist before the treatment decision. In Danilack, they are the variables used to predict induction.

Now comes the deepest idea.

Each subject has two potential outcomes.

One potential outcome is what would happen if that subject received treatment.

The other potential outcome is what would happen if that same subject received control.

For a pregnant patient in week thirty nine, one potential outcome is whether she would have a caesarean if induced that week. The other potential outcome is whether she would have a caesarean if managed expectantly that week.

The individual causal effect is the difference between those two potential outcomes.

But here is the fundamental problem of causal inference.

For any one patient, we can observe only one of those two outcomes.

If she is induced, we observe what happens after induction. We do not observe what would have happened to that same patient under expectant management.

If she is managed expectantly, we observe what happens after expectant management. We do not observe what would have happened to that same patient

under induction.

This missing counterfactual is the central problem.

Causal inference is the science of making disciplined comparisons despite the fact that each person reveals only one of their two possible worlds.

Now, since we cannot estimate the individual causal effect directly for each patient, we usually estimate average causal effects.

One target is the average treatment effect. Spoken plainly, this asks: in the whole population, what would be the average difference if everyone were treated compared with if everyone were untreated?

Another target is the average treatment effect on the treated. This asks: among the people who actually received treatment, what was the average difference between what happened under treatment and what would have happened if those same people had not received treatment?

These are not the same question.

The average treatment effect is a population-wide question.

The average treatment effect on the treated is a question about the treated patients.

This distinction matters in Danilack because propensity score matching often moves us toward an effect among treated, matchable patients. It tells us something like: among women who were actually induced and could be matched to similar expectantly managed women, what was the estimated effect of induction?

Now let us compare randomized trials and observational studies.

In a randomized trial, treatment assignment is independent of the patient's potential outcomes. In ordinary language, whether the patient gets treatment is not chosen because of hidden prognosis. Randomization breaks the link between clinical risk and treatment assignment.

This is why randomization is so powerful.

If treatment is randomized, then the treated and untreated groups should be comparable not only for measured baseline variables, but also for unmeasured ones, at least on average.

In an observational study, treatment assignment is not independent. It depends on patient features.

In Danilack, induction depends on clinical features. Some women are induced because the pregnancy has become medically riskier. Those same features may influence caesarean risk.

That is confounding by indication in mathematical form: treatment assignment depends on baseline covariates, and those baseline covariates are related to the

potential outcomes.

To estimate causal effects from observational data, we need assumptions.

The first major assumption is called unconfoundedness.

Unconfoundedness means this: after conditioning on the observed baseline covariates, treatment assignment is independent of the potential outcomes.

Let us translate.

If we knew and adjusted for all the relevant measured baseline variables, then whether a patient received treatment would be as-if random among patients with the same covariate profile.

In Danilack, this would mean that among women with the same measured clinical and demographic profile, whether one was induced and another was managed expectantly is not systematically related to what their caesarean outcomes would have been under either strategy.

That is a strong assumption.

It is not automatically true.

It depends on whether the important confounders were actually measured and correctly included.

The second major assumption is called positivity, or overlap.

Positivity means that every type of patient being studied must have some non-zero chance of receiving either treatment.

Spoken clinically, for each baseline profile, induction must be possible and expectant management must be possible.

If a certain kind of patient is always induced, then the data contain no comparable expectantly managed patients for that type. If another kind of patient is never induced, then the data contain no comparable induced patients for that type.

Without overlap, the study cannot learn the counterfactual from the data. It must extrapolate beyond what was observed.

This is not a technical nuisance. It is a causal boundary.

If the patients do not overlap, the comparison is not supported.

Now we can define the propensity score mathematically, but in words.

The propensity score is the probability of receiving treatment, given the observed baseline covariates.

In Danilack, it is the probability of induction in a specific gestational week, given the measured profile of that patient.

The true probability function is unknown. We estimate it from the data. Most commonly, this is done with logistic regression, where the outcome of that regression is treatment assignment, not the clinical outcome.

So, in the propensity model, induction is the dependent variable. The predictors are the baseline covariates.

The fitted model produces, for each patient, a predicted probability of induction. That predicted probability is the estimated propensity score.

Now let us discuss the two key theoretical properties.

The first is the balancing property.

The balancing property says that if we condition on the propensity score, the distribution of observed baseline covariates should no longer depend on treatment assignment.

That sentence is dense, so let us say it more slowly.

Among patients with the same propensity score, the treated and untreated groups should have similar distributions of the measured baseline variables.

If two groups both have a propensity score around zero point seven, then one group should not be dramatically older, sicker, more hypertensive, more diabetic, or more growth-restricted than the other, at least for variables included in the propensity model.

This is why the propensity score is called a balancing score.

It does not merely predict treatment. It creates a basis for balance.

The second property is even more remarkable.

If treatment assignment is unconfounded after conditioning on the full set of observed covariates, then treatment assignment is also unconfounded after conditioning only on the propensity score.

In plain language, if the full covariate list was enough to remove measured confounding, then the one-number propensity score is also enough to remove that measured confounding.

This is the theoretical miracle of propensity scores.

We do not need to match exactly on eighty variables. We can match, stratify, weight, or adjust using one scalar probability.

But do not misunderstand the miracle.

The theorem does not say that the propensity score handles unmeasured confounding.

It says that, under the assumption that the observed covariates are sufficient, the score can replace the full observed covariate vector for the purpose of balancing

treatment assignment.

That is powerful, but conditional.

Now let us make this concrete with a simplified Danilack example.

Suppose a thirty eight-week patient has gestational hypertension, no prior caesarean, singleton vertex pregnancy, valid parity data, and a hospital profile where induction is commonly used for similar patients. The propensity model estimates that her probability of induction at thirty eight weeks is zero point seventy five.

Another thirty eight-week patient is not induced, but based on her observed profile, her probability of induction is also around zero point seventy five.

These two patients may differ in many details, but they occupy a similar position in the treatment-selection process. Both looked like plausible induction candidates.

If we compare induced patients with non-induced patients in this same propensity-score region, we are no longer comparing all induced women with all non-induced women. We are comparing women who had similar measured chances of being induced.

That is the intuitive meaning of conditioning on the propensity score.

Now let us talk about the estimators again, this time more mathematically but still orally.

Matching estimates the treatment effect by constructing pairs or sets.

For each treated subject, the method searches for an untreated subject with a similar propensity score. Once the matched sample is formed, the outcome comparison is made inside that matched sample.

The mathematical idea is that the matched untreated patient serves as an observed substitute for the treated patient's missing counterfactual.

Of course, it is not literally the same patient. But if the covariates are balanced, the untreated match helps approximate what would have happened under control.

Stratification estimates the treatment effect by dividing the propensity-score scale into bands.

Within each band, treated and untreated patients should be more comparable. The treatment effect is estimated inside each band, and then the band-specific estimates are averaged.

This is less exact than matching, but it uses more of the sample.

Inverse probability of treatment weighting has a different mathematical flavor.

It creates a synthetic population.

Patients who received the treatment they were likely to receive get modest weights. Patients who received the treatment they were unlikely to receive get larger weights.

Why?

Because those unlikely-treatment patients are especially informative. They represent the kind of patient for whom the opposite treatment was possible but did not occur often.

In a weighted pseudo-population, the distribution of measured covariates should become similar between treated and untreated groups. Treatment assignment becomes independent of measured covariates, at least if the model is correct and the weights behave well.

This is why extreme weights are dangerous.

If a patient had a probability of treatment near zero but actually received treatment, the inverse probability weight becomes huge. That one patient can start to dominate the analysis. The estimate becomes unstable because the data are being stretched too far.

Covariate adjustment using the propensity score is mathematically simpler.

The outcome model includes treatment assignment and the propensity score. The score acts as a one-number adjustment for the baseline covariates that predicted treatment.

This can be useful, but it is not automatically the strongest method. A single linear adjustment for the propensity score may not fully balance covariates. Balance still needs to be checked.

Now let us talk about balance diagnostics.

The mathematical theorem gives us the target. But the data must be inspected.

After propensity score matching, stratification, or weighting, we ask: are the measured baseline covariates actually balanced?

The standardized difference is a common tool. It measures how far apart the treatment groups are on a baseline variable, scaled by the variability of that variable. Unlike a p value, it is not mainly driven by sample size. It tells us about the magnitude of imbalance.

A standardized difference below zero point one is often considered acceptable, though this is a convention rather than a law of nature.

The deeper point is that propensity score analysis must show its homework.

It is not enough to say, we used propensity scores.

We must ask: Did the score create overlap? Did matching or weighting produce covariate balance? Were important confounders measured before treatment? Were

unmatched patients excluded, and if so, who were they? Were extreme weights present? Which causal effect is being estimated?

Now let us return to the two assumptions, because they are the guardrails.

Unconfoundedness says: after measured covariates are accounted for, treatment assignment is as-if random with respect to the potential outcomes.

Positivity says: every patient type in the analysis has a real chance of receiving either treatment.

If unconfoundedness fails because an important predictor is unmeasured, the propensity score cannot fix it.

If positivity fails because the groups do not overlap, the propensity score cannot invent a comparison.

These failures are not cosmetic. They threaten the causal interpretation.

Now let us place Danilack inside this framework.

The treatment assignment is induction in a given gestational week.

The control condition is expectant management in that week.

The covariates are the measured baseline characteristics, including demographic factors, hospital, year, parity, common diagnosis codes, and obstetric conditions.

The two potential outcomes are caesarean status if induced that week, and caesarean status if managed expectantly that week.

The observed outcome is only one of those two. A woman either was induced or she was not. We do not observe both clinical paths in the same woman.

The causal target depends on the method.

The matched analysis is closest to an effect among women who were induced and could be matched to comparable expectantly managed women.

The propensity-score-adjusted regression is closer to a population-average association among women in the overlapping region of the propensity score.

This is why Danilack's paper carefully distinguishes matching from adjustment.

The two methods are not just computational variants. They are asking related but subtly different causal questions.

Now let us make the mathematical intuition memorable with one final image.

Imagine baseline covariates as a huge clinical map. Every patient sits somewhere on that map. The map has many axes: age, parity, gestational week, hypertensive disease, diabetes, growth restriction, hospital practice pattern, and many more.

Exact matching on the full map is nearly impossible. There are too many dimensions.

The propensity score draws a contour line through that map.

Patients on the same contour line may not be identical, but they have the same modeled probability of receiving treatment. They are similar in the way that matters for treatment assignment.

So the score lets us compare patients along the same treatment-selection contour.

That is what the mathematical theorem formalizes.

Conditional on that contour, the measured covariates should balance. And if the measured covariates were enough to remove confounding, then the contour is enough to support a causal comparison.

But the map must include the important landmarks.

If a major landmark is missing, such as cervical dilation or body mass index, the contour may be misleading.

If one group lives on a part of the map where the other group has no patients, there is no honest comparison there.

This is the mathematical humility behind the method.

Now let us close this mathematical layer.

The propensity score is the estimated probability of treatment, given observed baseline covariates.

Its purpose is not to predict the outcome. Its purpose is to balance the measured causes of treatment assignment.

The potential outcomes framework reminds us that each patient has two possible outcomes, but only one is observed.

The assumptions of unconfoundedness and positivity tell us when observational data can support causal inference.

The balancing property tells us why one score can replace a high-dimensional covariate profile for purposes of adjustment.

And the Danilack paper is an applied example: many measured indications for induction are compressed into a week-specific propensity score, and that score is used to compare induced women with expectantly managed women who looked similarly likely to be induced.

That is the mathematical core.

Propensity scores do not eliminate uncertainty. They discipline it.

Clinical synthesis for the OBGYN resident

Now let us bring everything back to the labour ward.

You are an OBGYN resident. You are not reading this paper in the abstract. You are reading it because, on a real shift, real patients ask real questions.

Will induction increase my chance of caesarean?

Should we induce now, or wait?

Is waiting safer?

Is induction a pathway to vaginal birth, or is it a pathway to surgery?

This paper teaches us that the answer cannot be reduced to the old slogan, induction increases caesareans.

That slogan is too crude.

It ignores the most important clinical fact: patients are induced for reasons.

A patient induced at thirty six weeks for pre-eclampsia is not the same as a patient in spontaneous labour at thirty six weeks without complications. A patient induced at thirty eight weeks for growth restriction is not the same as a patient who arrives in spontaneous labour with a normally grown fetus and reassuring physiology. A patient induced at forty one weeks with oligohydramnios, a borderline tracing, suspected macrosomia, or thick meconium risk is not the same as a patient who labours spontaneously at forty one weeks.

So the resident must learn to think in counterfactuals.

The clinical question is not: what happens after induction compared with spontaneous labour?

The clinical question is: what happens after induction compared with expectant management?

This distinction changes bedside reasoning.

If the alternative to induction is spontaneous labour in the next few hours in an uncomplicated patient, that is one clinical comparison.

If the alternative to induction is several more days of pregnancy in a patient with worsening hypertension, decreasing fluid, suspected placental insufficiency, or late-term risk, that is a very different comparison.

Expectant management is not nothing.

Waiting is an intervention in time.

Waiting carries its own risks. It may allow spontaneous labour. It may also allow the cervix to ripen, the fetus to grow, the placenta to age, fluid to decrease, hypertensive disease to worsen, diabetes-related risks to accumulate, or fetal reserve to narrow.

This is the clinical logic behind the Danilack study design. The comparator was not just non-induced labour. It was expectant management, week by week.

Now let us translate the paper's week-by-week results into clinical thinking.

Before forty weeks, after propensity score matching, induction did not appear to increase caesarean risk. In fact, the matched estimates from thirty four through thirty nine weeks were below one.

This does not mean that every induction before forty weeks is easy.

It does not mean that every patient should be induced.

It does not mean that caesarean risk disappears.

It means that, among women who were induced and compared with similar women managed expectantly, the act of induction itself did not look like the cause of excess caesareans before forty weeks.

For a resident, this is liberating but also sobering.

It is liberating because it challenges an overly fearful view of induction. Induction is not automatically a failure of physiologic birth. When there is a valid indication, induction may be the path that prevents deterioration and still achieves vaginal delivery.

It is sobering because induction remains a complex clinical process. The success of induction depends on cervical status, parity, fetal condition, contraction pattern, uterine response, patient preferences, staffing, monitoring, and patience with the labour curve.

The paper does not say induction is harmless. It says the crude blame placed on induction is often confounded.

Now think about the patient counseling conversation.

A weak counseling statement would be: induction increases the risk of caesarean.

A better counseling statement would be: older observational studies sometimes found more caesareans after induction, but many of those comparisons were biased because women are induced for medical reasons that already increase caesarean risk. When studies compare induction with expectant management and adjust for those reasons, induction before forty weeks does not necessarily increase caesarean risk.

For a patient at thirty nine weeks, you might say this in simpler language.

We do not compare induction with an imaginary perfect spontaneous labour. We compare induction with waiting. In many analyses, when patients are compared fairly with similar patients who waited, induction around this gestational age does not appear to raise caesarean risk, and may sometimes be associated with lower risk.

That is a clinically honest explanation.

Now consider the forty-week and forty one-week findings.

At forty and forty one weeks, Danilack found a moderately elevated caesarean risk after induction, even after propensity score matching.

How should a resident interpret that?

Do not overstate it.

It may represent a true increased caesarean risk with induction in later gestation. It may reflect residual confounding. It may be influenced by late-term biology, including larger fetal size, lower fluid, meconium, less fetal reserve, or a lower tolerance for prolonged induction. It may also be affected by how the comparison group is structured near the end of the dataset.

But do not ignore it either.

At forty and forty one weeks, the induction decision should be made with attention to cervical favorability, parity, fetal size, fetal testing, amniotic fluid, maternal comorbidity, and the department's ability to support a careful induction.

The point is not that forty-week induction is bad.

The point is that the later gestational-age setting is different. The balance between induction, waiting, fetal risk, and caesarean risk becomes more nuanced.

Now let us talk about cervical status, because it is clinically central and methodologically important.

The Danilack dataset lacked cervical dilation and body mass index. That matters.

At the bedside, you know why it matters. A nulliparous patient with a closed, posterior cervix is not the same as a multiparous patient at three centimeters. Both may be induced, but their induction trajectories are not the same.

Cervical favorability is not just a number. It is a marker of how close the reproductive system is to labour. An unfavorable cervix means the induction must first create cervical readiness before labour progress can occur. That takes time. It requires ripening, patience, and careful interpretation of failed induction criteria.

If a study cannot measure cervical status, it cannot fully separate the biological readiness for labour from the decision to induce.

This is one of the clinical places where residual confounding may live.

Now let us talk about parity.

Parity is one of the most important clinical modifiers of induction success. A multiparous uterus and cervix have already passed through labour. A nulliparous induction is often longer and more vulnerable to diagnosis of arrest, failed induction, or fetal intolerance over time.

Danilack included parity, which is important. But a resident must still individualize counseling. The paper gives a population-level estimate. The patient in front of you has a parity, cervix, indication, tracing, preferences, and clinical trajectory.

Now let us talk about fetal monitoring and time.

An induction creates a long observation window. During that window, fetal heart rate patterns may evolve. Tachysystole can occur. Decelerations can appear. Maternal fever, epidural effects, chorioamnionitis, exhaustion, or arrest disorders may enter the story.

But again, do not make the naive mistake.

Those events are not always caused by the induction. Some are caused by the underlying condition. Some are caused by the duration of labour. Some are revealed by labour. Some are consequences of uterine stimulation. Some are consequences of the clinical threshold for intervention.

The resident's job is to keep causal thinking alive during clinical work.

When a caesarean occurs after induction, ask what actually led to it.

Was it failed cervical ripening?

Was it fetal intolerance before active labour?

Was it tachysystole from oxytocin?

Was it arrest after adequate contractions?

Was it worsening pre-eclampsia?

Was it suspected macrosomia?

Was it a non-reassuring tracing in a fetus that may already have had reduced reserve?

Each pathway has a different lesson.

Now let us translate the paper into practical resident behavior.

First, when presenting an induction plan, always state the indication and the alternative.

Do not say only, we are inducing for gestational hypertension.

Say, we are inducing for gestational hypertension at thirty seven weeks, and the alternative is expectant management with continued maternal and fetal monitoring, which carries risk of disease progression.

This forces the team to compare induction with the correct comparator.

Second, separate baseline risk from intervention risk.

If the patient has pre-eclampsia, diabetes, fetal growth restriction, oligohydramnios, or late-term pregnancy, those factors carry risk before the first dose of cervical

ripening agent is given. Do not attribute all later caesarean risk to induction itself.

Third, counsel in probabilities, not absolutes.

Say that induction may be longer, especially with an unfavorable cervix, but modern evidence comparing induction with expectant management does not support a simple claim that induction inevitably increases caesarean risk before forty weeks.

Fourth, document the reasoning.

A good induction note should make the counterfactual visible. It should explain why induction is being chosen now, what risks are being avoided by not waiting, and what factors may affect success.

Fifth, manage induction with patience when clinically safe.

One way induction can become a caesarean risk is through impatience. If the maternal and fetal status are reassuring, the cervix is still ripening, membranes are intact or recently ruptured, and adequate time has not passed, premature diagnosis of failed induction can create avoidable caesareans.

The paper does not directly study labour management protocols, but the clinical implication is clear. If induction itself is not necessarily the enemy, then the quality of induction management matters.

Now let us move to the clinician-researcher view.

Research methods synthesis

As a clinician-researcher, the first thing to admire in this paper is the question design.

The authors did not fall into the common trap of comparing induction with spontaneous labour as though spontaneous labour were the only alternative. They compared induction with expectant management. That is clinically coherent.

When a clinician decides whether to induce at thirty eight weeks, the alternative is not spontaneous labour at thirty eight weeks. The alternative is to wait. The patient may then deliver at thirty eight, thirty nine, forty, or later weeks, by spontaneous labour, later induction, or caesarean. That is the real clinical decision structure.

This is one of the strongest features of the paper.

The second strength is the week-specific analysis.

Gestational age is not just a covariate. It changes the entire clinical context. Induction at thirty four weeks is usually medically driven. Induction at thirty nine weeks may include elective or soft indications. Induction at forty one weeks reflects late-term physiology and practice patterns.

By analyzing thirty four through forty two weeks separately, the authors respect the changing clinical meaning of induction across gestation.

The third strength is the large sample.

A cohort of more than one hundred sixty six thousand linked mother-newborn records allows week-specific estimates and broad propensity score models. This is exactly the type of setting where propensity scores can be useful: many measured confounders, a large dataset, and a treatment-selection process that must be modeled carefully.

The fourth strength is the non-parsimonious propensity score model.

In ordinary prediction modeling, parsimony can be attractive. But in propensity score modeling, the goal is not elegant prediction for its own sake. The goal is covariate balance. Including many clinically relevant baseline variables can be appropriate, even if some variables are not individually statistically significant.

The authors understood this. They attempted to model the treatment-selection process broadly.

The fifth strength is that they compared analytic approaches.

They showed crude estimates, traditionally adjusted estimates, propensity-score-matched estimates, and propensity-score-adjusted estimates. This allows the reader to see how the association moves as confounding control becomes more sophisticated.

That movement is itself informative.

When crude estimates are higher, traditional adjustment moves them downward, and propensity score matching moves them further downward, the pattern suggests that baseline differences were materially affecting the crude association.

Now let us critique the study.

The first limitation is observational design.

No matter how sophisticated the analysis, this is not a randomized trial. Propensity scores balance measured covariates. They do not guarantee balance of unmeasured covariates.

The second limitation is administrative data.

Administrative codes can be powerful at scale, but they are blunt. They may misclassify indications, diagnoses, onset of labour, induction procedures, and caesarean context. Coding accuracy can vary across hospitals and over time.

In obstetrics, timing is everything. A diagnosis before induction is a confounder. A diagnosis after induction may be a mediator or consequence. Administrative data often struggle to preserve this temporal sequence.

The third limitation is missing clinical granularity.

Body mass index and cervical dilation were not available. These are not minor omissions. They are clinically meaningful predictors of induction course and caesarean risk.

Body mass index affects labour progress, induction duration, fetal size risk, anesthesia complexity, and caesarean likelihood.

Cervical dilation and overall Bishop score affect whether induction begins near labour or far from labour.

If these variables differ between induced and expectantly managed patients, residual confounding may remain.

The fourth limitation is unmeasured clinician judgment.

Clinicians induce not only because of coded diagnoses, but because of a total impression. Fetal testing may be borderline but not coded with precision. Fluid may be low-normal. Growth may be concerning but not below a strict threshold. The patient may live far away. There may be prior obstetric history not fully captured. The cervix may be favorable. The department may have local practice norms.

Some of this lives outside the dataset.

The fifth limitation is changing matchability across gestational age.

The match rate was excellent at thirty four through thirty eight weeks, lower at thirty nine through forty one weeks, and much lower at forty two weeks. That tells us something about overlap.

At forty two weeks especially, only about fifty eight percent of inductions were matched. This means the matched estimate applies to a restricted subset. The unmatched patients may be clinically important.

The sixth limitation is the late-gestation comparator.

At forty two weeks, the expectant management group is structurally limited because there is little later pregnancy left in the dataset. This can distort the comparison. The authors explicitly acknowledge the issue of truncation at forty two weeks.

The seventh limitation is that propensity score adjustment and propensity score matching estimate different things.

This is not a flaw, but it is a point the reader must understand. If the estimates differ, we should not ask only which method is right. We should ask which causal question each method is answering, and whether the assumptions are plausible.

Now let us ask what would make the evidence stronger.

A randomized trial can solve many confounding problems, but trials are expensive, selective, and not always feasible for every gestational week and indication.

A stronger observational study would include richer clinical data: Bishop score, body mass index, exact indication, fetal testing, estimated fetal weight, amniotic fluid, cervical ripening method, oxytocin duration, membrane status, time from induction start to active labour, and standardized criteria for failed induction and arrest.

It would also separate elective induction from medically indicated induction, nulliparous from multiparous patients, favorable from unfavorable cervix, and different gestational-age windows.

It would examine not only caesarean delivery, but also maternal morbidity, postpartum hemorrhage, infection, neonatal intensive care admission, shoulder dystocia, stillbirth, hypertensive progression, patient experience, and length of stay.

Why?

Because caesarean delivery is not the only outcome that matters.

A resident must not reduce obstetrics to caesarean avoidance alone. The goal is not vaginal birth at any cost. The goal is the best maternal and neonatal outcome aligned with patient values.

Now let us identify the paper's main research-methods lesson.

The methods teach us that a clinical question must be framed around the real decision.

Induction versus spontaneous labour is often the wrong question.

Induction versus expectant management is usually the right question.

The second methods lesson is that adjustment changes meaning.

Crude estimates answer a crude question. Traditional adjustment answers a somewhat better question. Propensity matching answers a question about comparable, matchable patients. Propensity adjustment answers a more model-based population question.

When estimates differ, the difference is not noise. It is a clue about confounding and about the target population.

The third methods lesson is that balance is more important than the mere label of the method.

A paper that says propensity score analysis but does not show overlap, match quality, or covariate balance has not earned causal confidence.

Danilack did examine balance, match rates, and unbalanced covariates. That strengthens interpretability.

The fourth methods lesson is humility.

A well-done observational study can move us closer to causal understanding. It cannot fully erase the fact that treatment was not randomized.

Now let us wrap the entire lecture.

The Danilack paper begins with a common clinical fear: induction may increase caesarean delivery.

The paper then shows that the crude association is heavily shaped by confounding by indication. Women are induced for reasons, and those reasons often carry caesarean risk.

The authors use propensity scores to model the probability of induction, week by week, based on measured baseline characteristics. This allows a fairer comparison between induced women and expectantly managed women with similar measured likelihoods of induction.

The central result is that before forty weeks, propensity-score-matched induction did not appear to increase primary caesarean risk. From thirty four through thirty nine weeks, the matched risk ratios were below one. At forty and forty one weeks, induction was associated with a moderate increase in caesarean risk. At forty two weeks, interpretation was limited by sample size and overlap.

The clinical lesson is not that everyone should be induced.

The clinical lesson is that induction should be judged against the correct alternative: expectant management in a patient with a specific gestational age, indication, cervix, parity, fetal condition, and maternal context.

The methodological lesson is that observational obstetric research must respect treatment selection. If sicker or higher-risk patients receive the intervention, crude outcome comparisons can punish the intervention for the risk carried by the indication.

The causal inference lesson is that propensity scores help compare patients who looked similarly likely to receive treatment, reducing confounding from measured variables.

The humility lesson is that propensity scores cannot balance what was not measured, and cannot create overlap where clinical practice provides none.

For the resident on labour ward, the final practical message is this.

Do not fear induction as a reflex.

Do not offer induction casually.

Do not compare induction with an imaginary spontaneous labour.

Compare induction with waiting.

State the indication. Assess the cervix. Consider parity. Think about fetal reserve. Counsel honestly. Manage patiently. Reassess continuously.

And when you read research, ask the same question you ask at the bedside.

Compared with what?

That question is the bridge between clinical medicine and causal inference.

It is also the bridge that makes this paper worth remembering.